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● ANSWER KEY

Advanced Math

04

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ANSWER KEY · SAT QUESTION BANK

2026-05-29

Q1**20**

The correct answer is 20. It's given that $fx = -a^x + b$. Substituting $-a^x + b$ for fx in the equation $y = fx - 12$ yields $y = -a^x + b - 12$. It's given that the y -intercept of the graph of $y = fx - 12$ is 0, $-\frac{75}{7}$. Substituting 0 for x and $-\frac{75}{7}$ for y in the equation $y = -a^x + b - 12$ yields $-\frac{75}{7} = -a^0 + b - 12$, which is equivalent to $-\frac{75}{7} = -1 + b - 12$, or $-\frac{75}{7} = b - 13$. Adding 13 to both sides of this equation yields $\frac{16}{7} = b$. It's given that the product of a and b is $\frac{320}{7}$, or $ab = \frac{320}{7}$. Substituting $\frac{16}{7}$ for b in this equation yields $a\frac{16}{7} = \frac{320}{7}$. Dividing both sides of this equation by $\frac{16}{7}$ yields $a = 20$.

Q2**17.5**

Accept also: 35/2

The correct answer is $\frac{35}{2}$. Subtracting the second equation from the first equation yields $y + k - y - k = x + 26 - x^2 - 5x$, or $2k = -x^2 + 6x + 26$. This is equivalent to $x^2 - 6x + 2k - 26 = 0$. It's given that the system has exactly one distinct real solution; therefore, this equation has exactly one distinct real solution. An equation of the form $ax^2 + bx + c = 0$, where a , b , and c are constants, has exactly one distinct real solution when the discriminant, $b^2 - 4ac$, is equal to 0. The equation $x^2 - 6x + (2k - 26) = 0$ is of this form, where $a = 1$, $b = -6$, and $c = 2k - 26$. Substituting these values into the discriminant, $b^2 - 4ac$, yields $-6^2 - 4(1)(2k - 26)$. Setting the discriminant equal to 0 yields $-6^2 - 4(1)(2k - 26) = 0$, or $-8k + 140 = 0$. Subtracting 140 from both sides of this equation yields $-8k = -140$. Dividing both sides of this equation by -8 yields $k = \frac{35}{2}$. Note that 35/2 and 17.5 are examples of ways to enter a correct answer.

Q3**.2916**

Accept also: .2917, 7/24

The correct answer is $\frac{7}{24}$. An expression of the form $\sqrt[n]{a^m}$, where m and n are integers greater than 1 and $a \geq 0$, is equivalent to $a^{\frac{m}{n}}$. Therefore, the expression on the right-hand side of the given equation, $\sqrt[3]{4^7}$, is equivalent to $4^{\frac{7}{3}}$. Thus, $4^{8c} = 4^{\frac{7}{3}}$. It follows that $8c = \frac{7}{3}$. Dividing both sides of this equation by 8 yields $c = \frac{7}{24}$. Note that 7/24, .2916, .2917, 0.219, and 0.292 are examples of ways to enter a correct answer.

Q4**76**

The correct answer is 76. It's given that the graph of $y = gx$ is the result of translating the graph of $y = fx$ up 4 units in the xy -plane. It follows that the graph of $y = gx$ is the same as the graph of $y = fx + 4$. Substituting gx for y in the equation $y = fx + 4$ yields $gx = fx + 4$. It's given that $fx = x - 6x - 2x + 6$. Substituting $x - 6x - 2x + 6$ for fx in the equation $gx = fx + 4$ yields $gx = x - 6x - 2x + 6 + 4$. Substituting 0 for x in this equation yields $g0 = 0 - 60 - 20 + 6 + 4$, or $g0 = 76$. Thus, the value of $g0$ is 76.

Q5**29/3**

Accept also: 9.666, 9.667

The correct answer is $\frac{29}{3}$. Applying the distributive property to the left-hand side of the given equation, $xx + 1 - 56$, yields $x^2 + x - 56$. Applying the distributive property to the right-hand side of the given equation, $4xx - 7$, yields $4x^2 - 28x$. Thus, the equation becomes $x^2 + x - 56 = 4x^2 - 28x$. Combining like terms on the left- and right-hand sides of this equation yields $0 = 4x^2 - x^2 + -28x - x + 56$, or $3x^2 - 29x + 56 = 0$. For a quadratic equation in the form $ax^2 + bx + c = 0$, where a , b , and c are constants, the quadratic formula gives the solutions to the equation in the form $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$. Substituting 3 for a , -29 for b , and 56 for c from the equation $3x^2 - 29x + 56 = 0$ into the quadratic formula yields $x = \frac{29 \pm \sqrt{(-29)^2 - 4(3)(56)}}{2(3)}$, or $x = \frac{29}{6} \pm \frac{13}{6}$. It follows that the solutions to the given equation are $\frac{29}{6} + \frac{13}{6}$ and $\frac{29}{6} - \frac{13}{6}$. Adding these two solutions gives the sum of the solutions: $\frac{29}{6} + \frac{13}{6} + \frac{29}{6} - \frac{13}{6}$, which is equivalent to $\frac{29}{6} + \frac{29}{6}$, or $\frac{29}{3}$. Note that 29/3, 9.666, and 9.667 are examples of ways to enter a correct answer.

Q6**C**C. $a^2 + 2ac + c^2$

Choice C is correct. Using the property that $\sqrt{x}\sqrt{y} = \sqrt{xy}$ for positive numbers x and y , with $x = (a + c)^3$ and $y = a + c$, it follows that $\sqrt{(a+c)^3} \cdot \sqrt{a+c} = \sqrt{(a+c)^4}$. By rewriting $(a + c)^4$ as $((a + c)^2)^2$, it is possible to simplify the square root expression as follows:
 $\sqrt{((a+c)^2)^2} = (a+c)^2 = a^2 + 2ac + c^2$.

Choice A is incorrect and may be the result of $\sqrt{(a+c)^3} \div \sqrt{(a+c)}$. Choice B is incorrect and may be the result of incorrectly rewriting $(a + c)^2$ as $a^2 + c^2$. Choice D is incorrect and may be the result of incorrectly applying properties of exponents.

Q7**-13/2**

Accept also: -6.5

The correct answer is $-\frac{13}{2}$. The value of x for which fx reaches its minimum can be found by rewriting the given equation in the form $fx = x - h^2 + k$, where fx reaches its minimum, k , when the value of x is h . The given equation, $fx = x - 2x + 15$, can be rewritten as $fx = x^2 + 13x - 30$. By completing the square, this equation can be rewritten as $fx = x^2 + 13x + \frac{13^2}{2} - 30 - \frac{13^2}{2}$, which is equivalent to $fx = x + \frac{13^2}{2} - \frac{289}{4}$, or $fx = x - \frac{13^2}{2} - \frac{289}{4}$. Therefore, fx reaches its minimum when the value of x is $-\frac{13}{2}$. Note that $-13/2$ and -6.5 are examples of ways to enter a correct answer.

Alternate approach: The graph of $y = fx$ in the xy -plane is a parabola. The value of x for the vertex of a parabola is the x -value of the midpoint between the two x -intercepts of the parabola. Since it's given that $fx = x - 2x + 15$, it follows that the two x -intercepts of the graph of $y = fx$ in the xy -plane occur when $x = 2$ and $x = -15$, or at the points $2, 0$ and $-15, 0$. The midpoint between two points, x_1, y_1 and x_2, y_2 , is $\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}$. Therefore, the midpoint between $2, 0$ and $-15, 0$ is $\frac{2 + (-15)}{2}, \frac{0 + 0}{2}$, or $-\frac{13}{2}, 0$. It follows that fx reaches its minimum when the value of x is $-\frac{13}{2}$. Note that $-13/2$ and -6.5 are examples of ways to enter a correct answer.

Q8**-3**

The correct answer is -3 . Squaring both sides of the given equation yields $x - 2^2 = 3x + 34$, which can be rewritten as $x^2 - 4x + 4 = 3x + 34$. Subtracting $3x$ and 34 from both sides of this equation yields $x^2 - 7x - 30 = 0$. This quadratic equation can be rewritten as $x - 10x + 3 = 0$. According to the zero product property, $x - 10x + 3$ equals zero when either $x - 10 = 0$ or $x + 3 = 0$. Solving each of these equations for x yields $x = 10$ or $x = -3$. Therefore, the given equation has two solutions, 10 and -3 . Of these two solutions, -3 is the smallest solution to the given equation.

Q9**8**

The correct answer is 8. Since each term of the given expression, $2x^3 + 42x^2 + 208x$, has a factor of $2x$, the expression can be rewritten as $2xx^2 + 2x21x + 2x104$, or $2xx^2 + 21x + 104$. Since the values 8 and 13 have a sum of 21 and a product of 104, the expression $x^2 + 21x + 104$ can be factored as $x + 8x + 13$. Therefore, the given expression can be factored as $2xx + 8x + 13$. It follows that the factors of the given expression are 2, x , $x + 8$, and $x + 13$. Of these factors, only $x + 8$ and $x + 13$ are of the form $x + b$, where b is a positive constant. Therefore, the possible values of b are 8 and 13. Thus, the smallest possible value of b is 8.

Q10**D**

D. Neither I nor II

Choice D is correct. A y -intercept of a graph in the xy -plane is a point where the graph intersects the y -axis, or a point where $x = 0$. Substituting 0 for x in the equation defining function f yields $f(0) = a2.2^0 + 2.2^b$, or $f(0) = a1 + 2.2^b$. So, the y -coordinate of the y -intercept of the graph is $a1 + 2.2^b$, or equivalently, $a + a2.2^b$. It's given that function g is equivalent to function f , where $0 < a < b$. It follows that $k = 2.2^b$. Since $a2.2^b$ can't be equal to 0, the coefficient a can't be equal to $a + a2.2^b$. Since $0 < a$, the constant k , which is equal to 2.2^b , can't be equal to $a + a2.2^b$. Therefore, function g doesn't display the y -coordinate of the y -intercept of the graph of $y = fx$ in the xy -plane as a constant or coefficient. It's also given that function h is equivalent to function f , where $0 < a < b$. The equation defining f can be rewritten as $fx = a2.2^x + a2.2^b$. It follows that $m = a2.2^b$. Since $a2.2^b$ can't be equal to 0, the coefficient a can't be equal to $a + a2.2^b$. Since $0 < a$, the constant m , which is equal to $a2.2^b$, can't be equal to $a + a2.2^b$. Therefore, function h doesn't display the y -coordinate of the y -intercept of the graph of $y = fx$ in the xy -plane as a constant or coefficient. Thus, neither function g nor function h displays the y -coordinate of the y -intercept of the graph of $y = fx$ in the xy -plane as a constant or coefficient.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Notes

Accept equivalent forms (e.g. $0.25 \equiv 1/4$). For grid-in items, decimal and fraction answers are both valid.

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